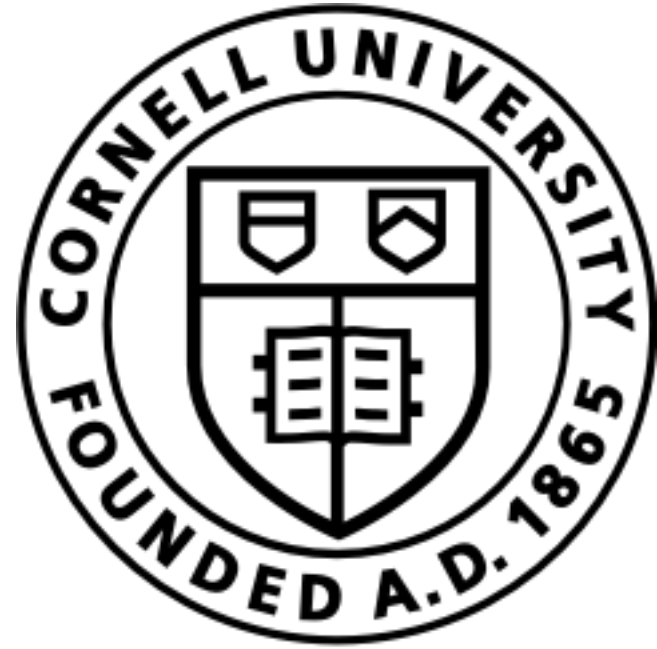




Lecture 1a: Introduction to course policies, procedures, and expectations. Motivating the study of decision-making.

CHEME 5760 Cornell University

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Agenda: In this lecture we will:

- ▶ **O1:** Introduce the course resources, policies, procedures, and expectations
- ▶ **O2:** Motivate why we should be interested in quantitative decision-making and better decisions (from the molecular to the population scale)
- ▶ **O3:** Introduce one of the key concepts that we will use for the rest of the course: **Utility and Utility functions**



Jeffrey Varner. BS/MS/PhD from Purdue University. Postdoc at ETHZ.
Worked at DuPont in Palo Alto, CA, before joining Cornell in 2005



I make **horrible** decisions—the worst. But I'm good at modeling and optimization. Can I turn decisions into optimization problems?



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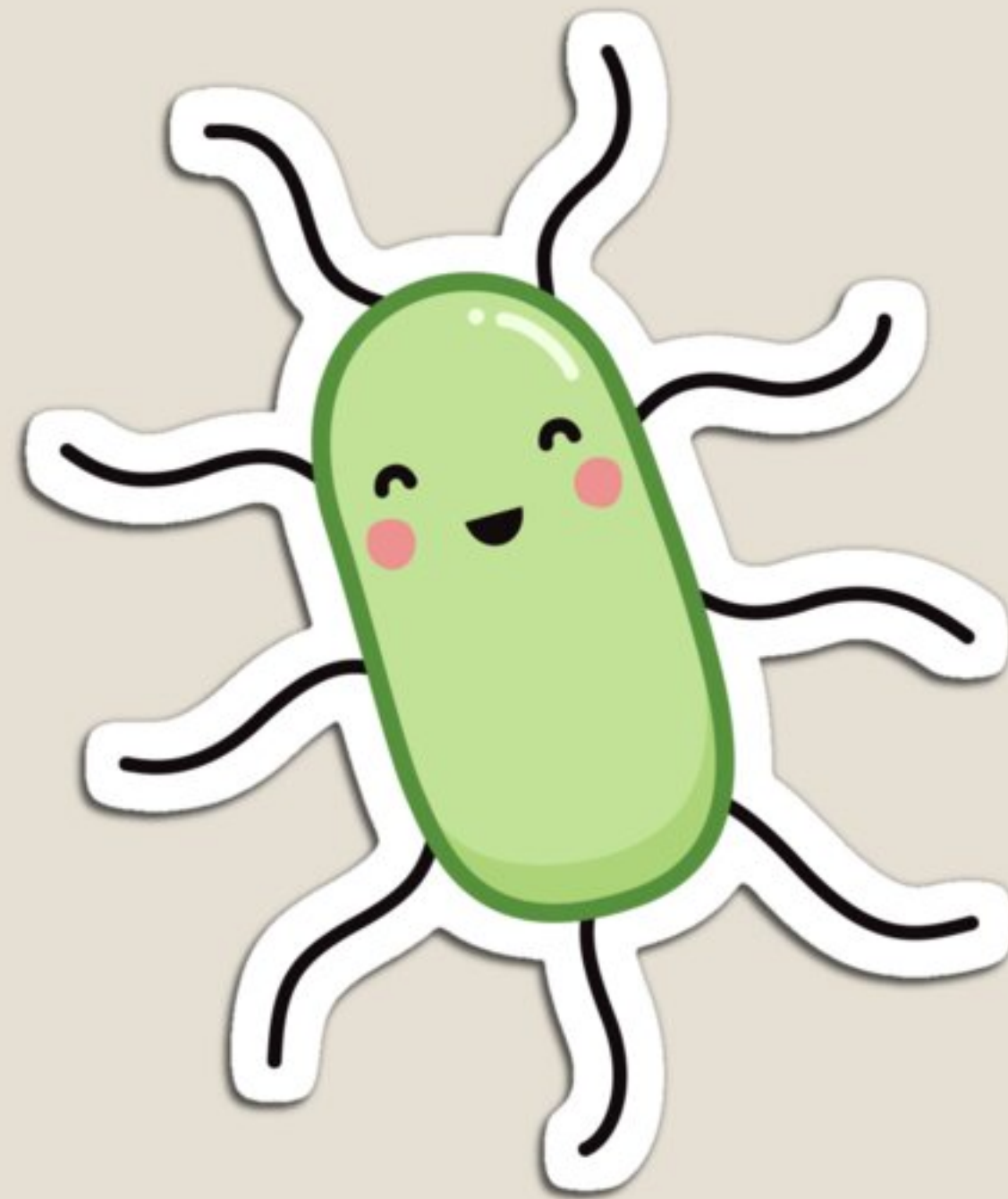


I make **horrible** decisions—the worst. But I'm good at modeling and optimization. Can I turn decisions into optimization problems?

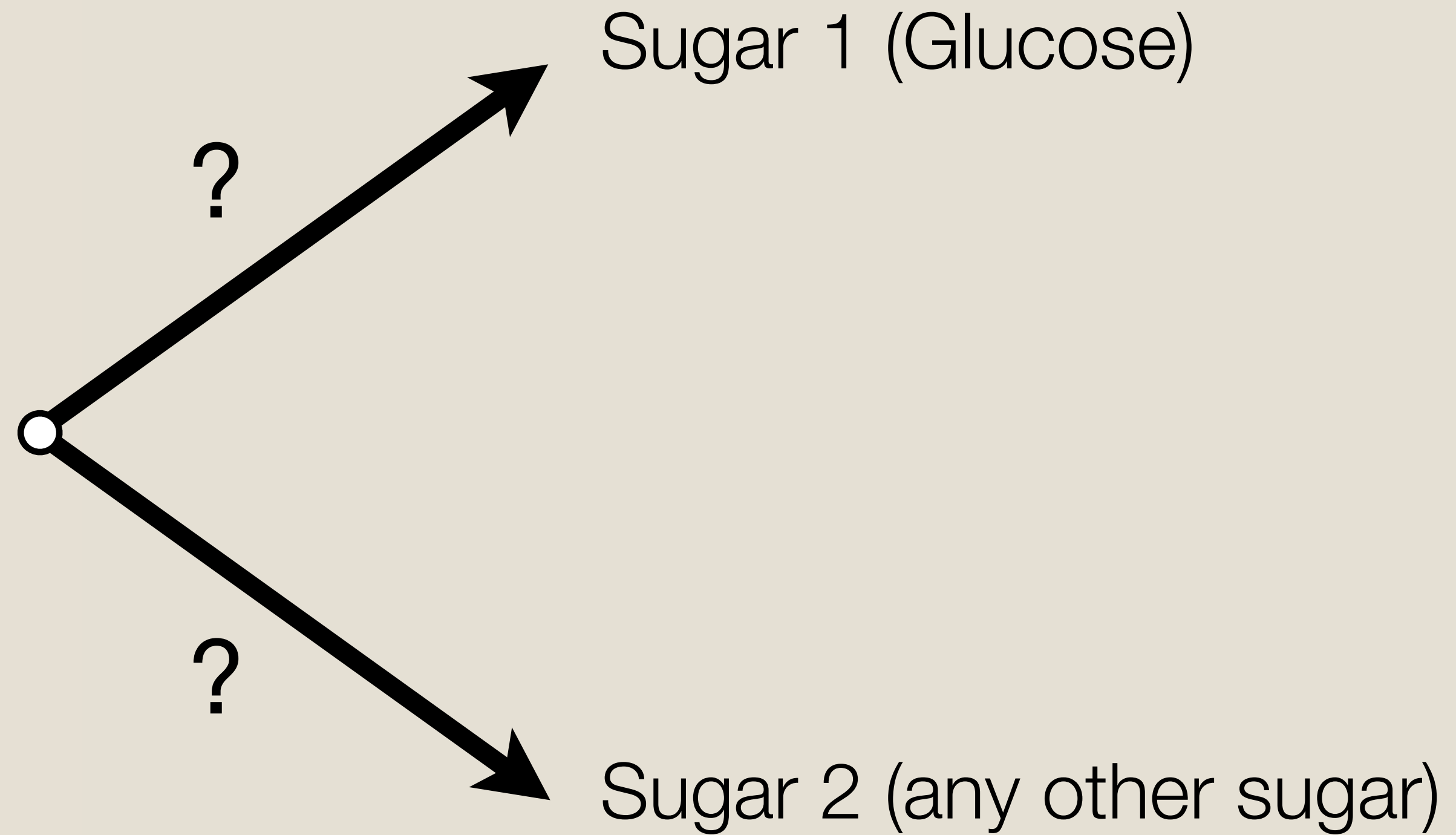


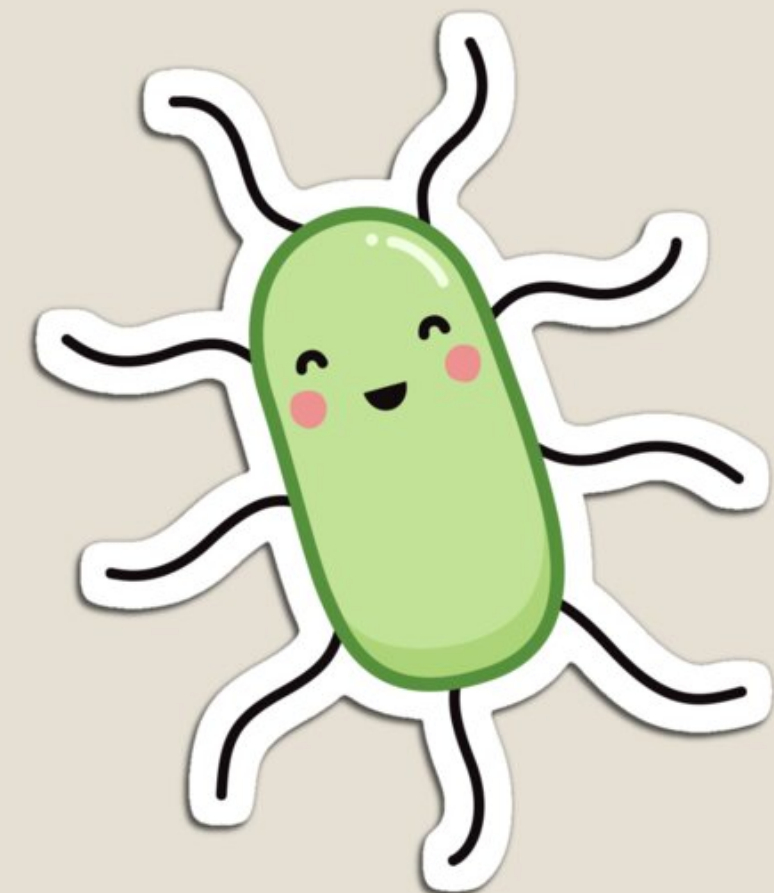
Course information: Policies, expectations, and materials for the course are hosted online and can be found at: <http://bit.ly/3KKdhdg>

Decisions at the: **Microscale**

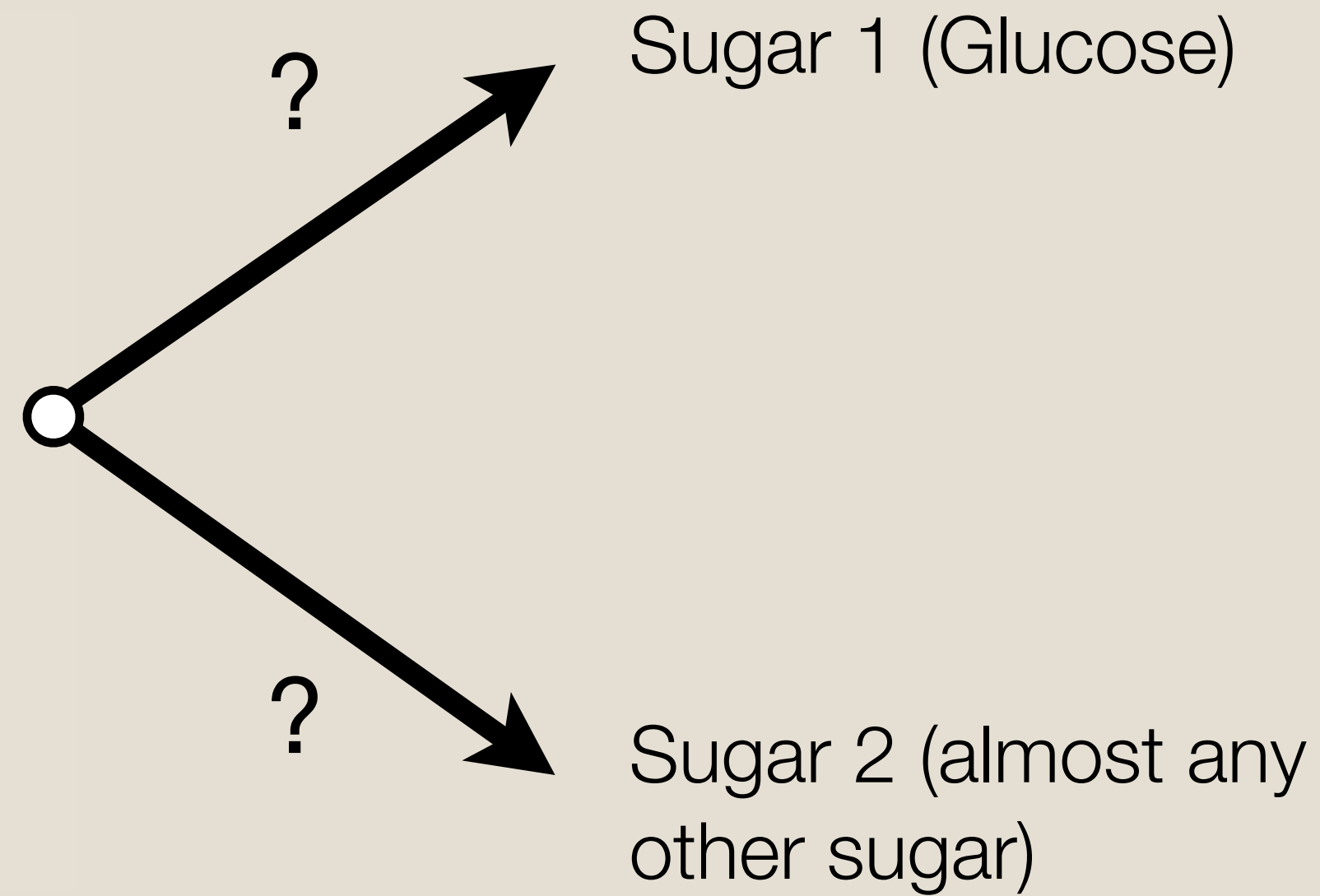


Escherichia coli

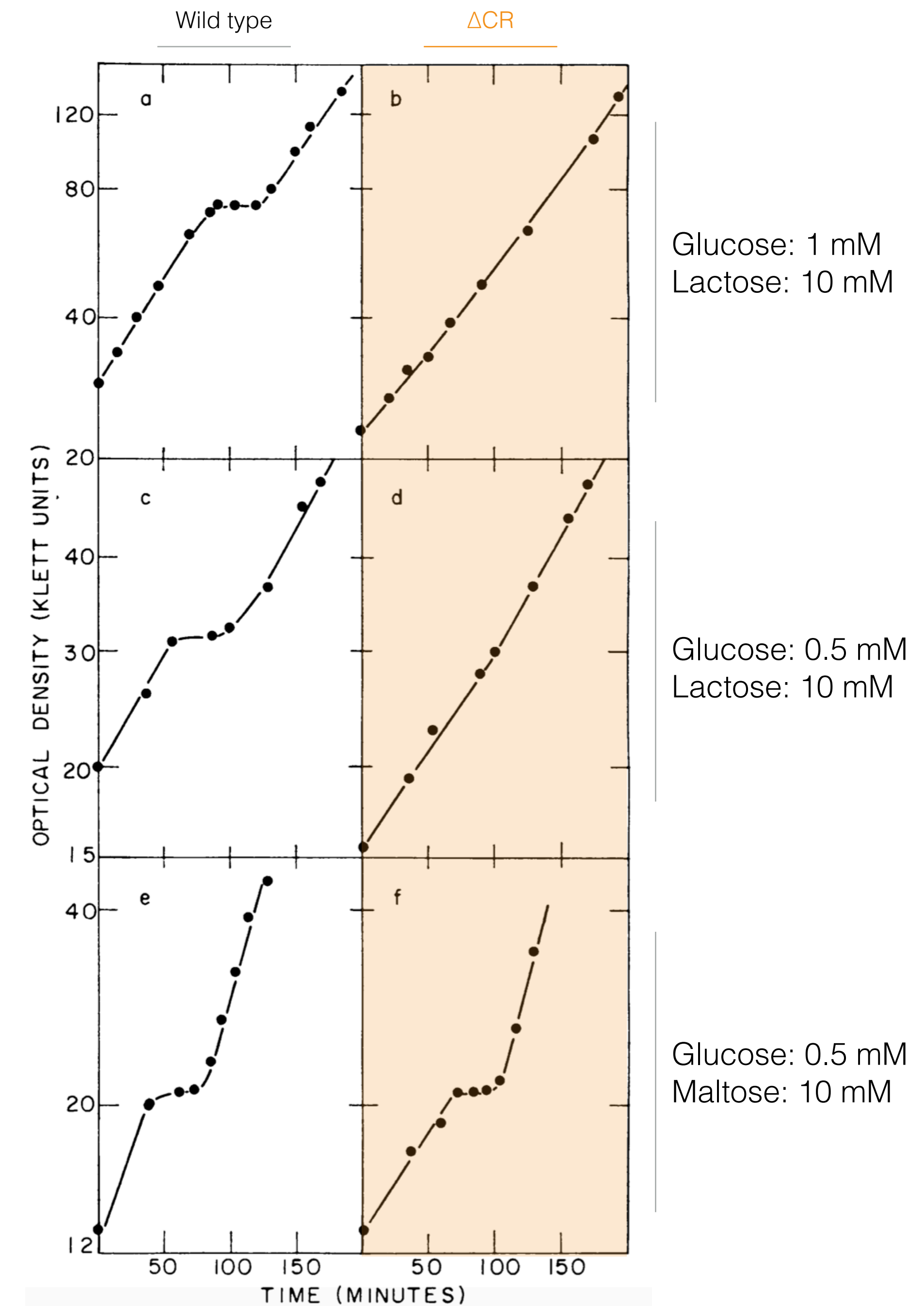




Escherichia coli



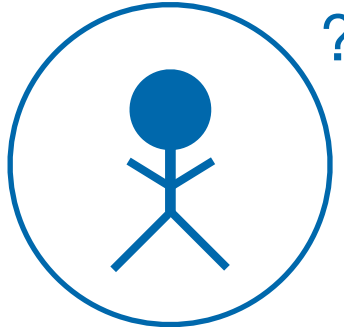
Loomis and Magasanik (1967) Glucose-Lactose
Diauxie in *Escherichia coli*. J. Bacteriol.,
93:1397-1401



Decisions at the: Peoplescale

Investment decision problems

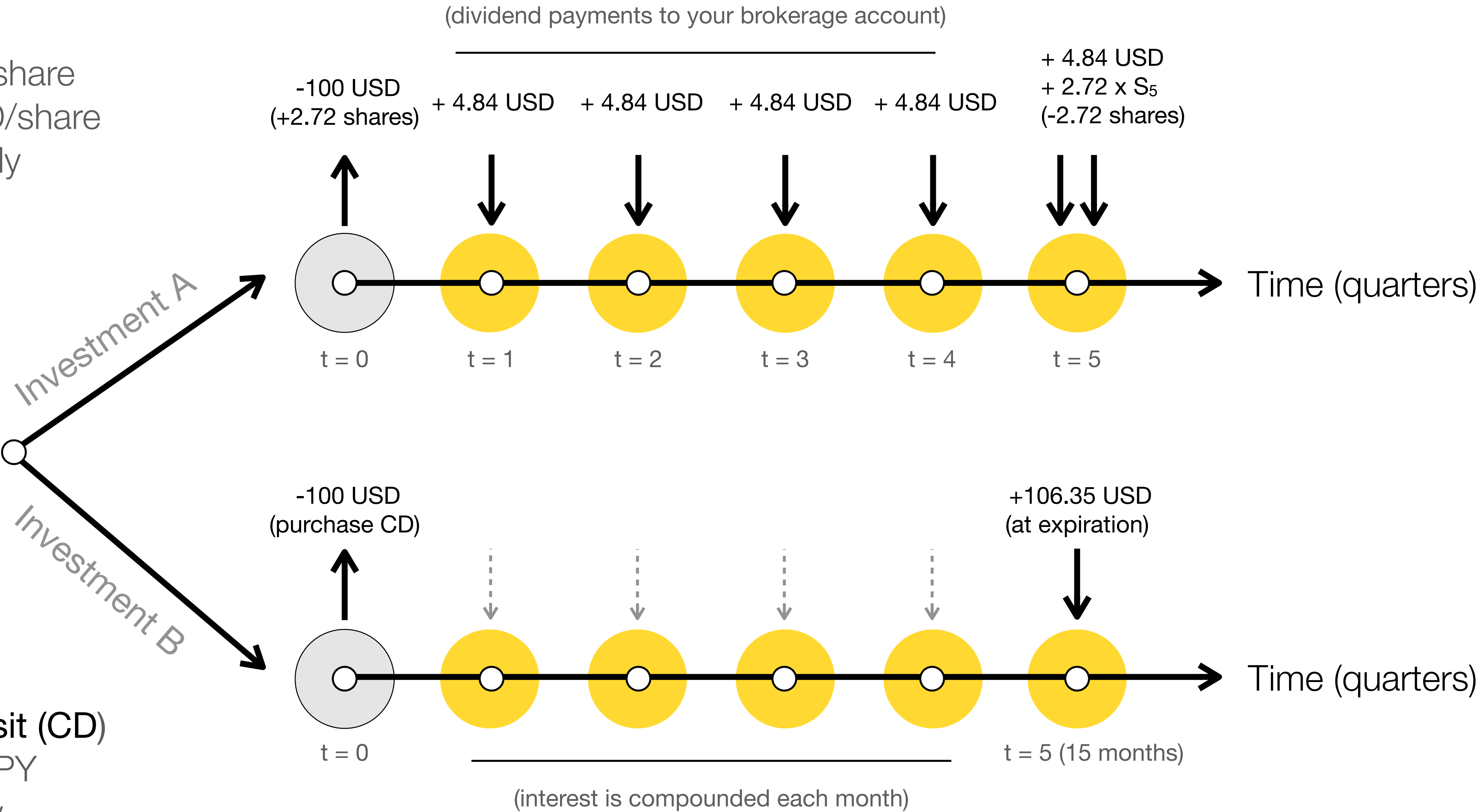
SPYD: 36.77 USD/share
Dividend: 1.78 USD/share
Frequency: Quarterly



100 USD

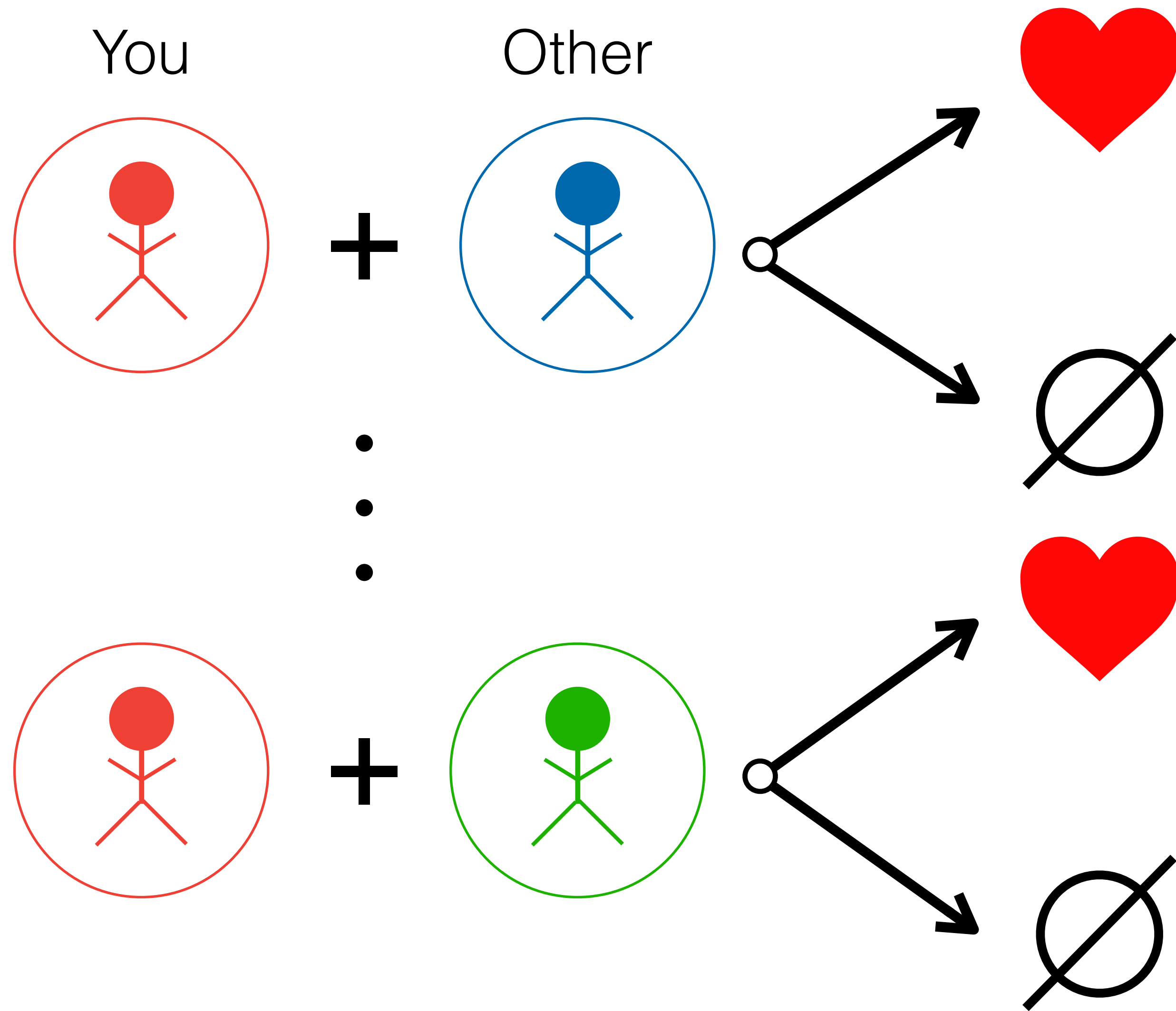
Perspective: Investor

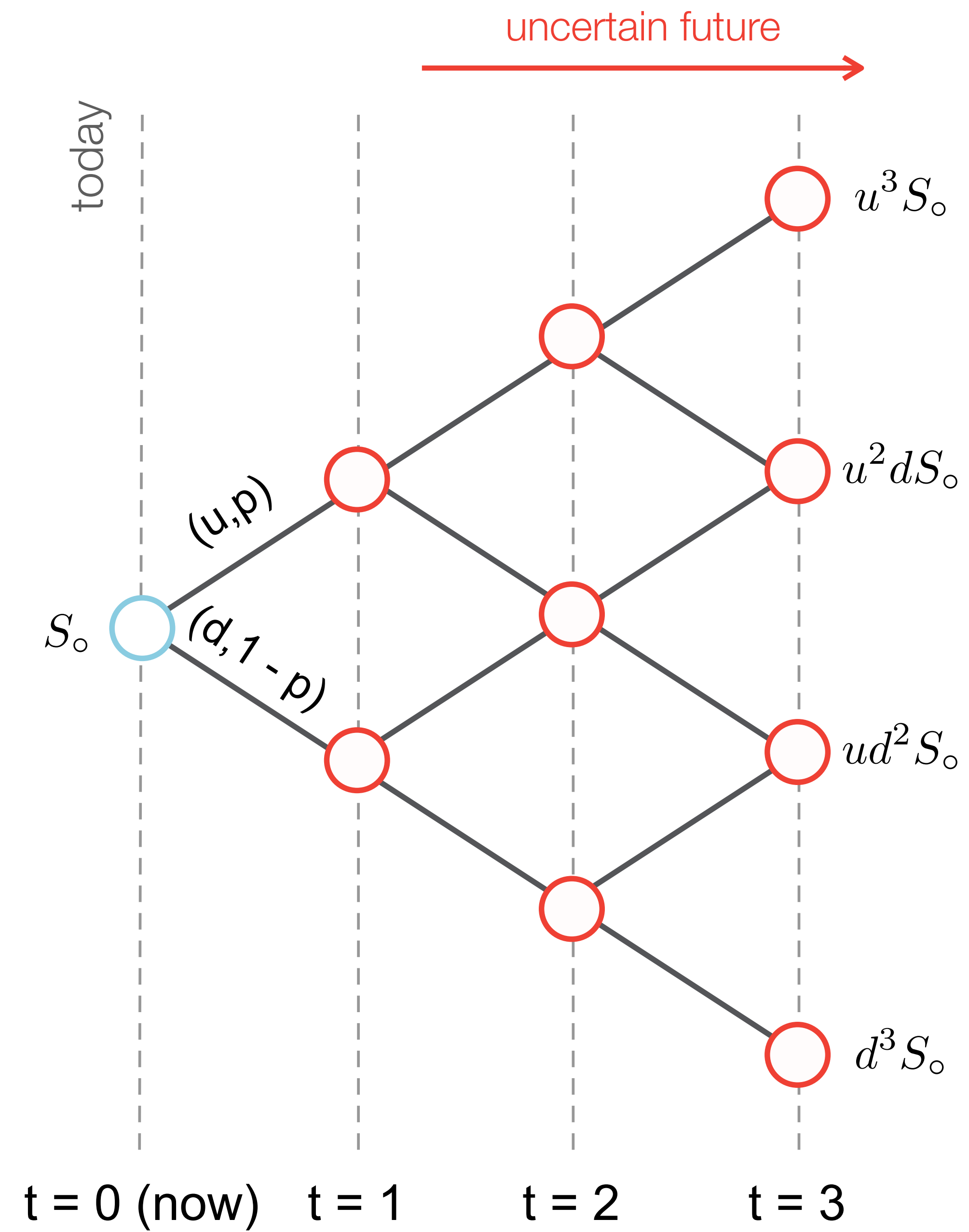
Certificate of deposit (CD)
Interest rate: 5% APY
Frequency: Monthly



Optimal stopping problems

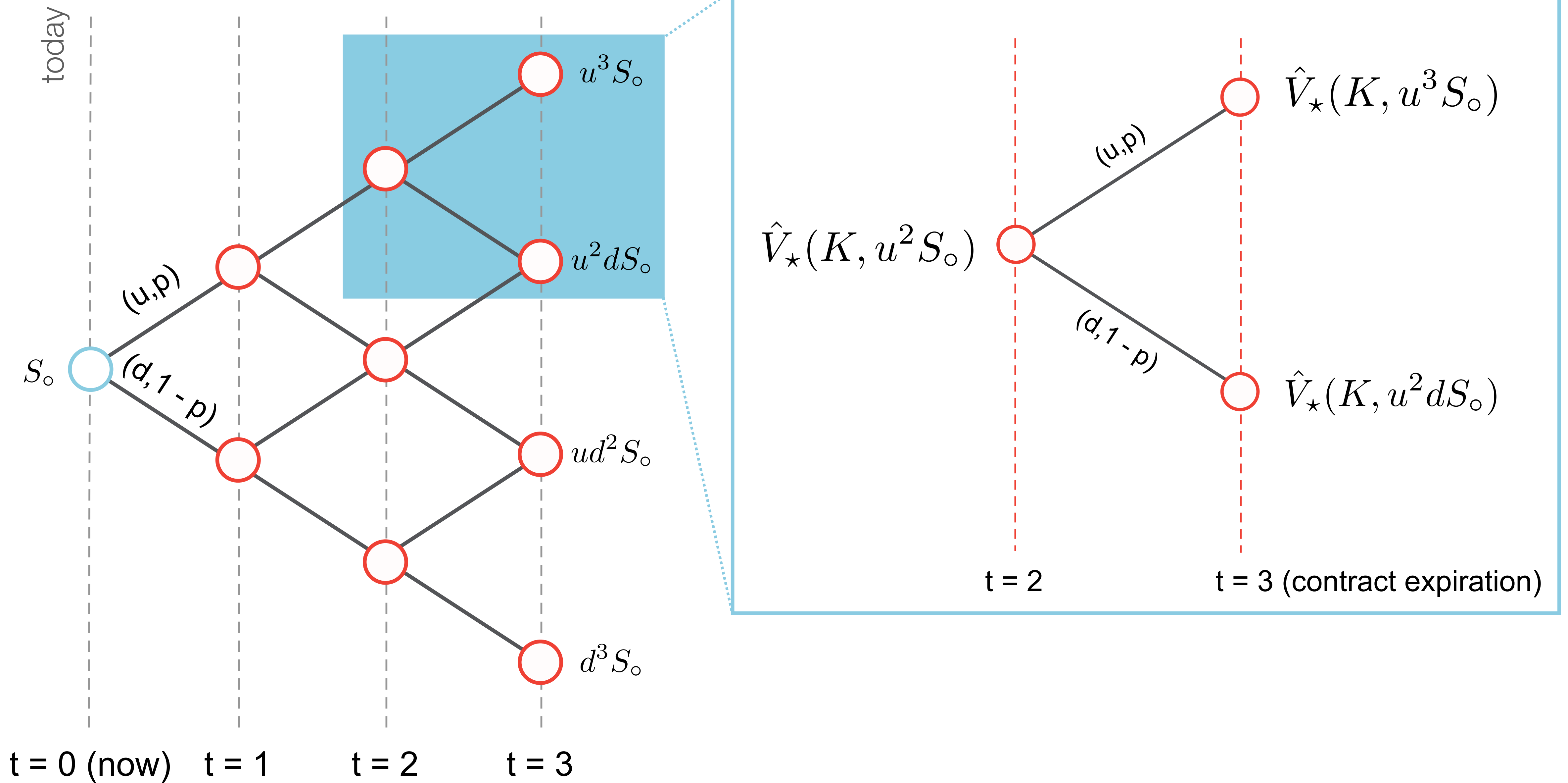
How many people do I
need to date?





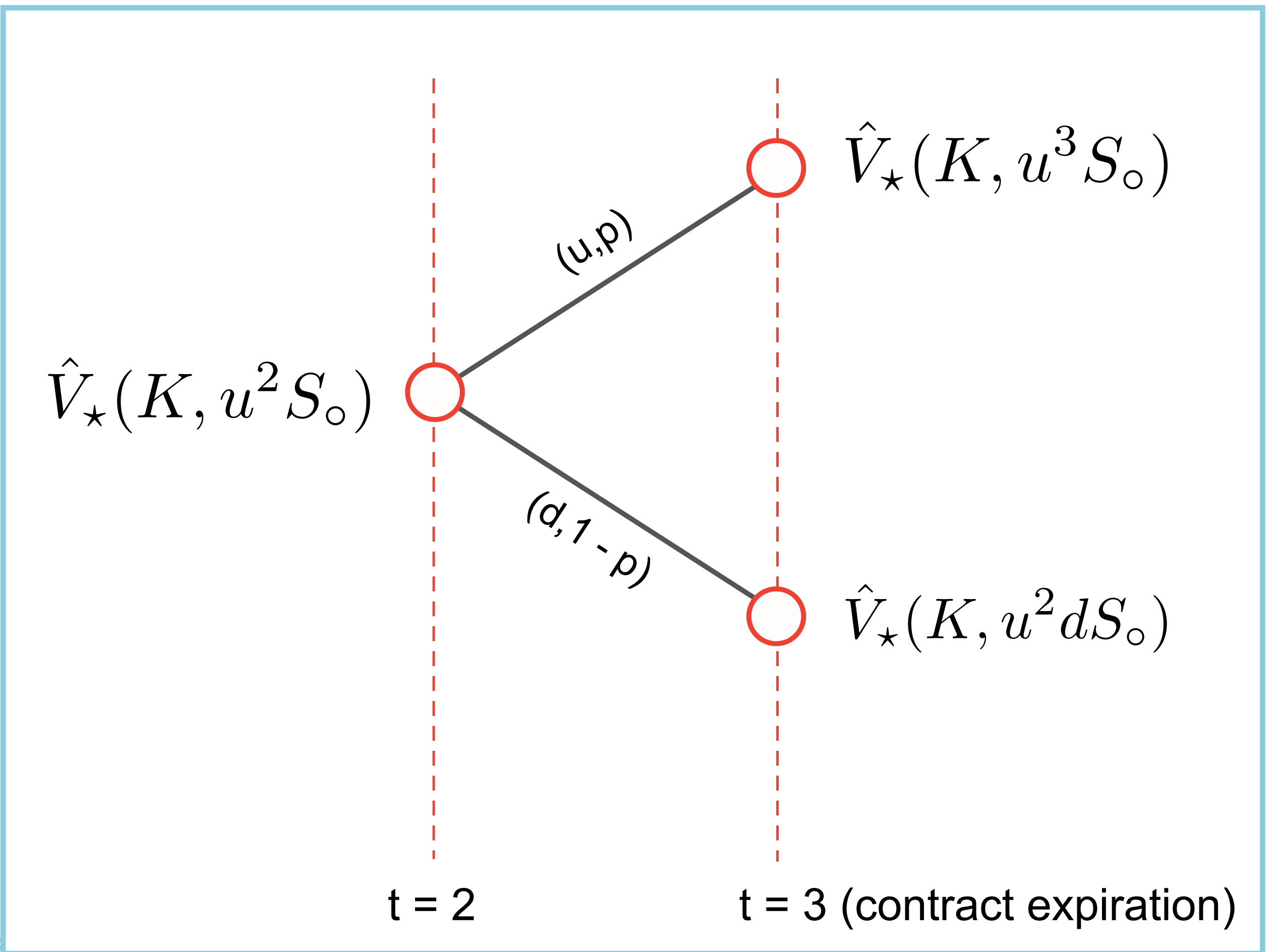
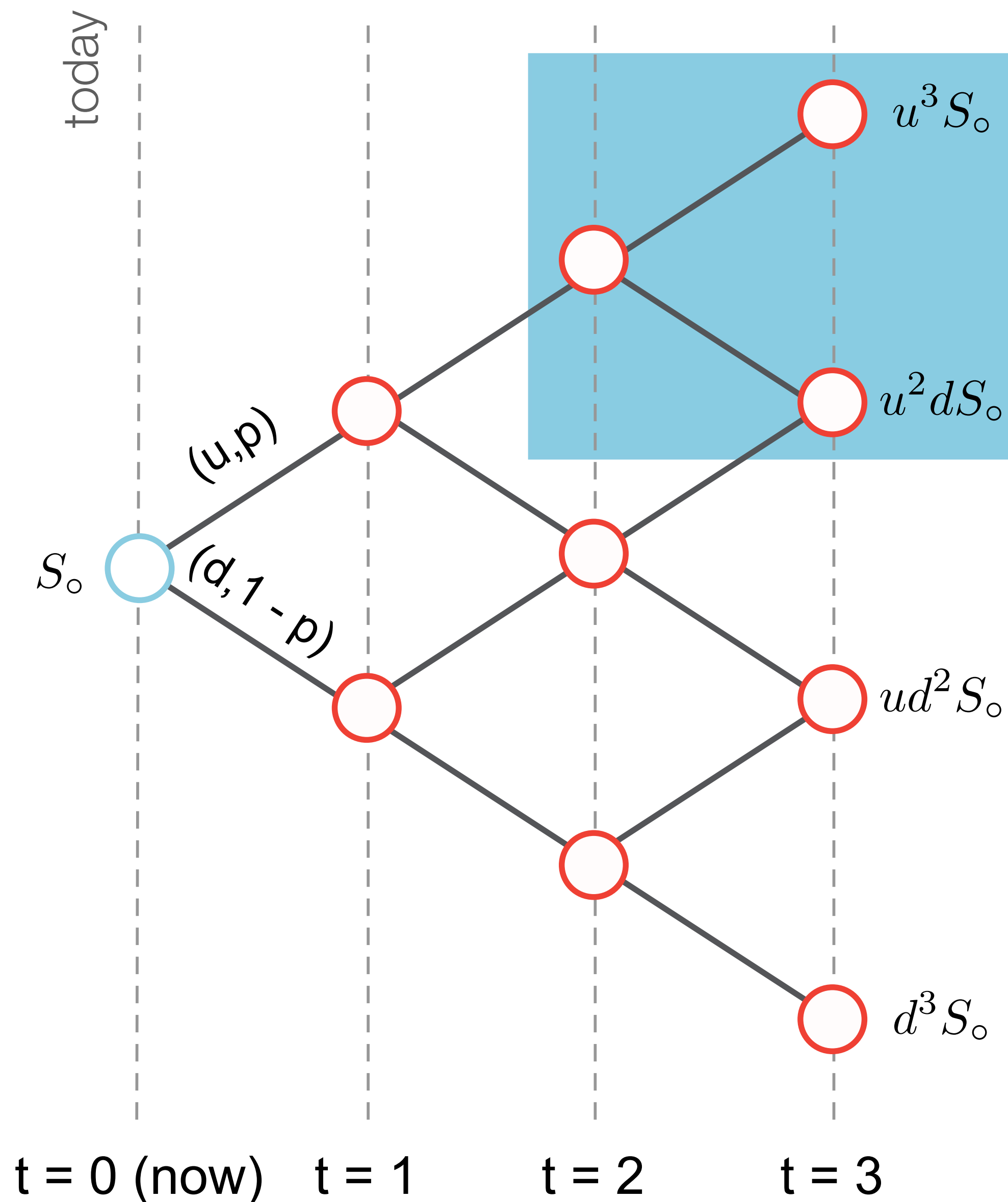
uncertain future

Single decision cell



uncertain future

Single decision cell



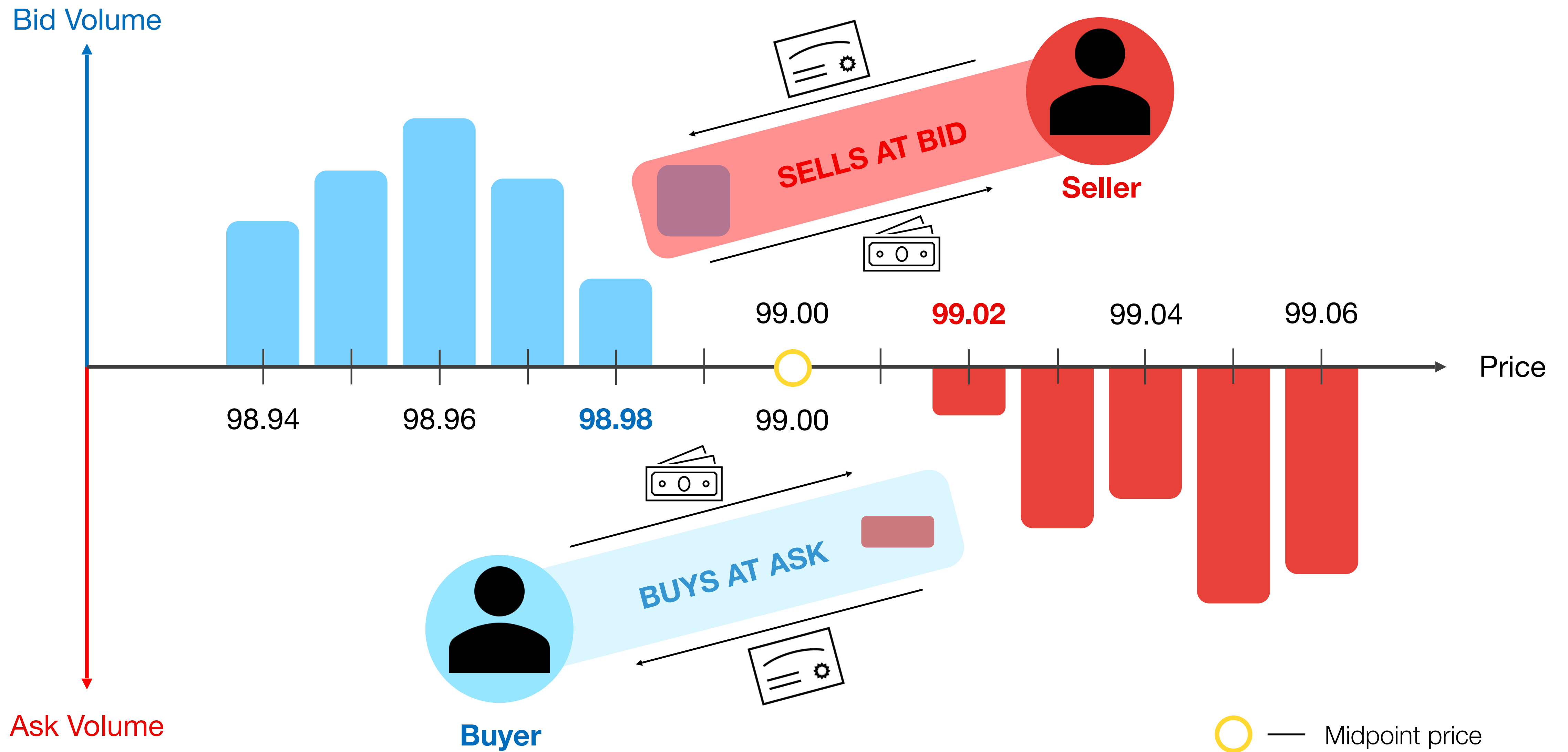
Decision rule American contract:

$$\hat{V}_{\star,t} = \max \left\{ \hat{V}_{\star}(K, u^2 S_0), \mathcal{D}_{t,t+1}^{-1} \mathbb{E} \left(\hat{V}_{\star,t+1} \right) \right\}$$

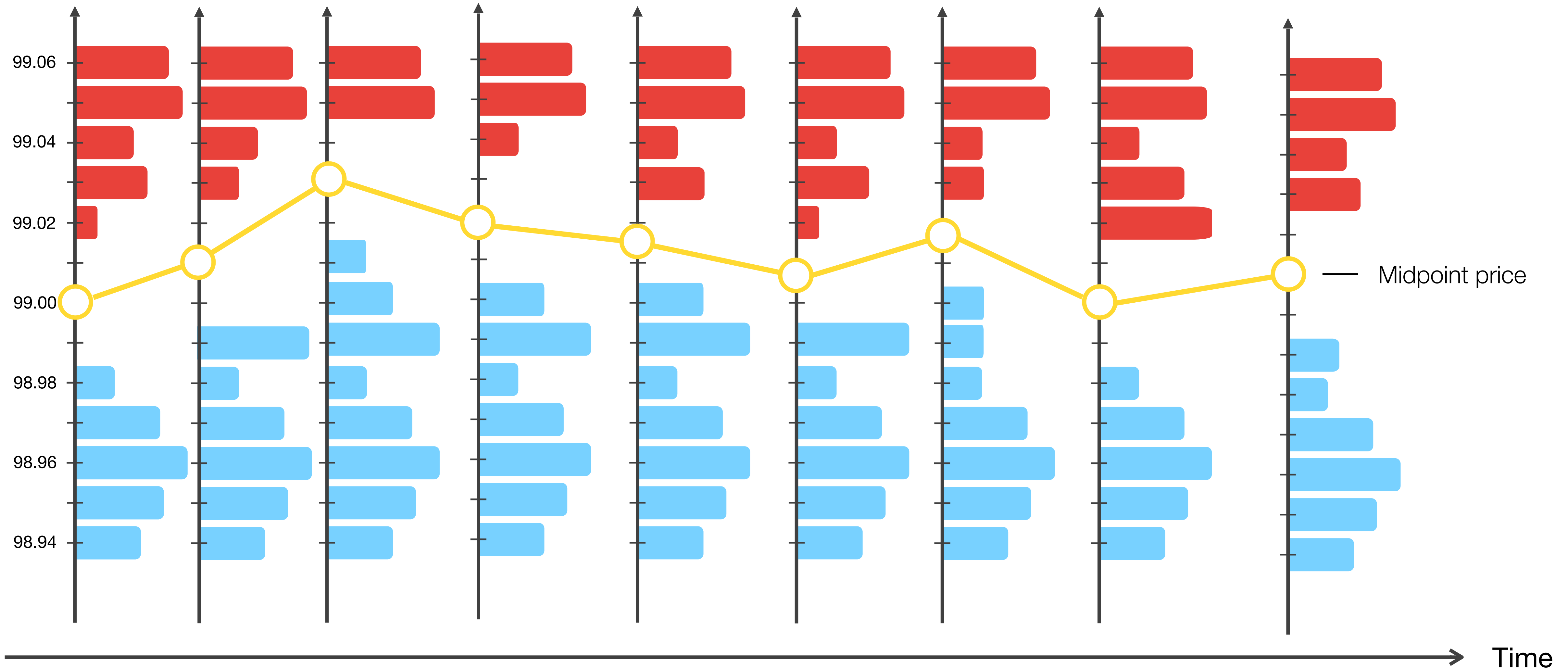
Exercise now

hold for the future

Decisions at the: Macroscale games

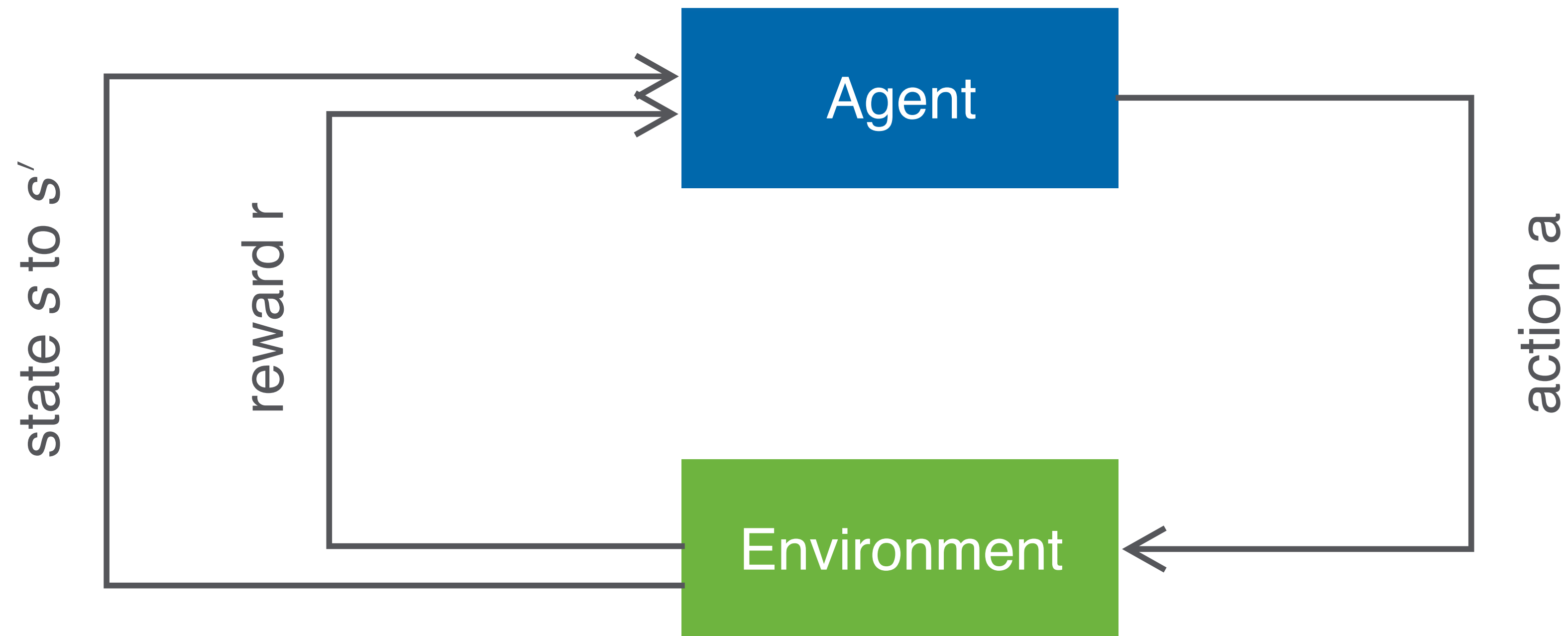


Emergent phenomena: Prices at double auctions emerge as the collective result of individual decisions



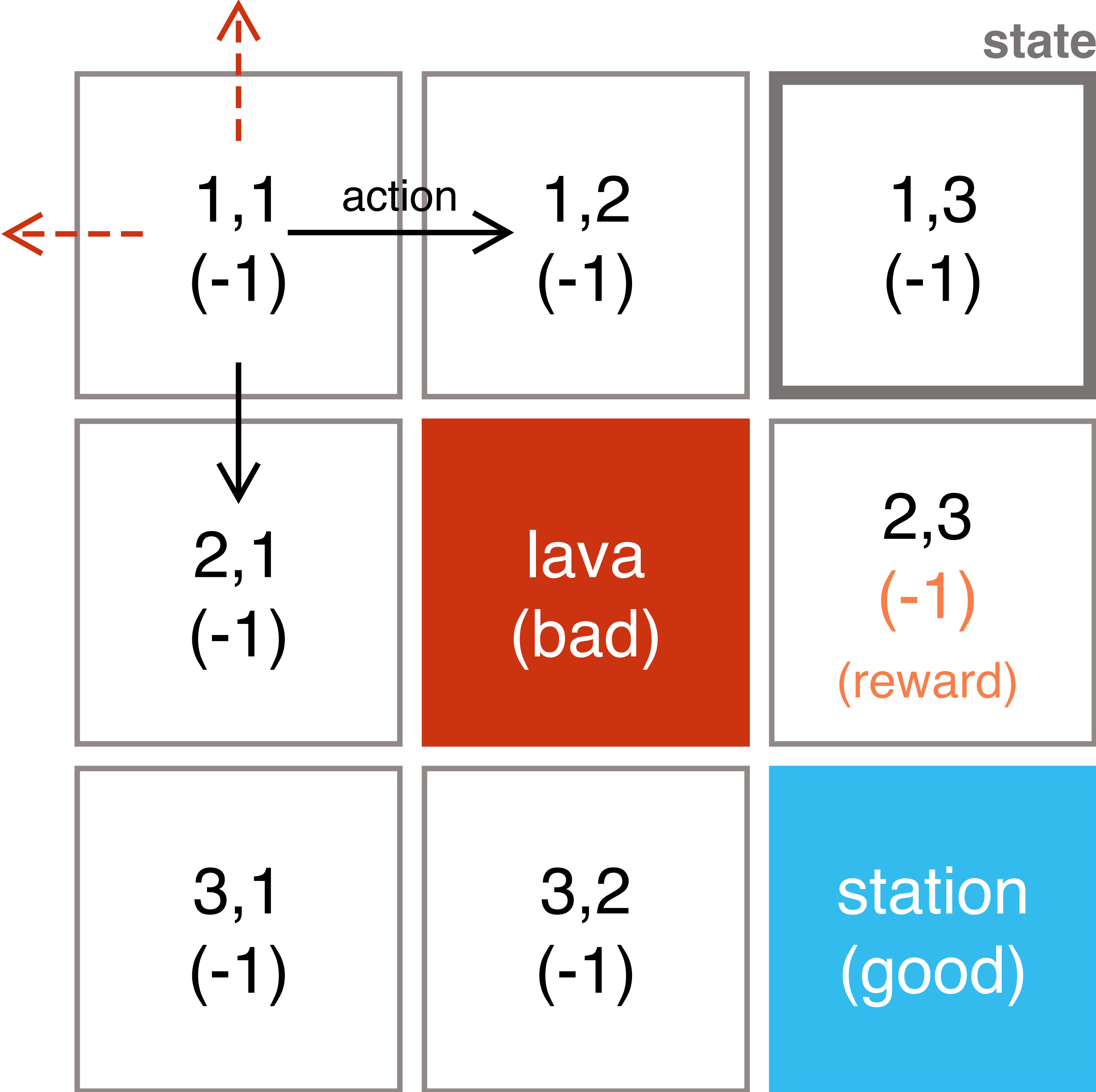
Learning: Learn by Doing

Decision-making agents use **reinforcement learning (RL)** to learn by doing (like humans do).



Agents select an action and watch the response of the world. They get a reward if outcome is good or a penalty if the outcome is bad

Grid world problems



- ▶ **Reinforcement learning** is studied in many disciplines, including game theory, control theory, operations research, information theory, simulation-based optimization, multi-agent systems, swarm intelligence, and statistics.
- ▶ **Model-based reinforcement learning** uses a *predictive model* of the world and rewards to ask questions of the form “What will happen if I do x?” to choose the best value for x
 - Sampling-based planning methods select candidate actions from a distribution, evaluates the samples using the model, and choose the most promising action.
- ▶ **Model-free reinforcement learning** bypasses the modeling step. Agents learn an optimal or near-optimal policy directly from *experimenting in real-time*.
 - Model-free approaches such as Q-learning or the closely related SARSA approach incrementally learn the state-action-value $Q(s, a)$ table from online experimentation. They learn by doing.

Shared idea throughout: Utility and
Utility Functions

Big idea for next time:

An individual decision-maker (agent) is given a set of n objects, $X = \{x_1, \dots, x_n\}$. A utility function ranks the agent's preference for combinations of these objects:

$$U(x_1, x_2, \dots, x_n) = u \tag{1}$$

where u is a real number called the utility which has units of utils.

- The utility function $U : X \rightarrow \mathbb{R}$ is unique only up to an order-preserving transformation.
- Utility functions are ordinal, i.e., they rank-order bundles but do not measure differences between bundles.

Today?